Homework 08

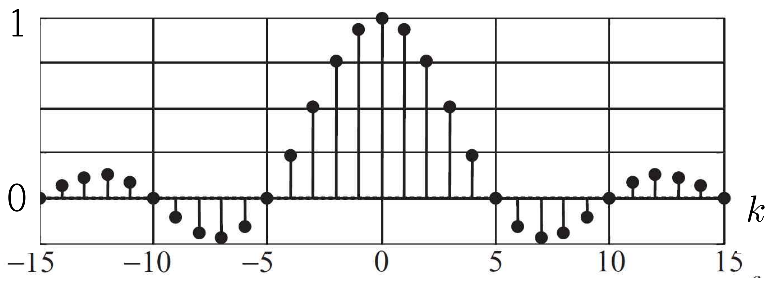
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

[1-9] Let a_k represent the Fourier series coefficients of the following periodic square wave. Are the following statements True [T] or False [F]?

1. a_k is real-valued.
2. $a_k = 0$ if k is even.
3. There are an infinite number of non-zero a_k.
4. There are an infinite number of zero a_k.
5. $a_0 = 0.5$.
6. The duty cycle of the square wave is 50 %.
7. The fundamental frequency is 400 Hz .
8. The first zero-crossing frequency is 2000π [rad/sec].
9. When the period of the periodic square wave is infinite,

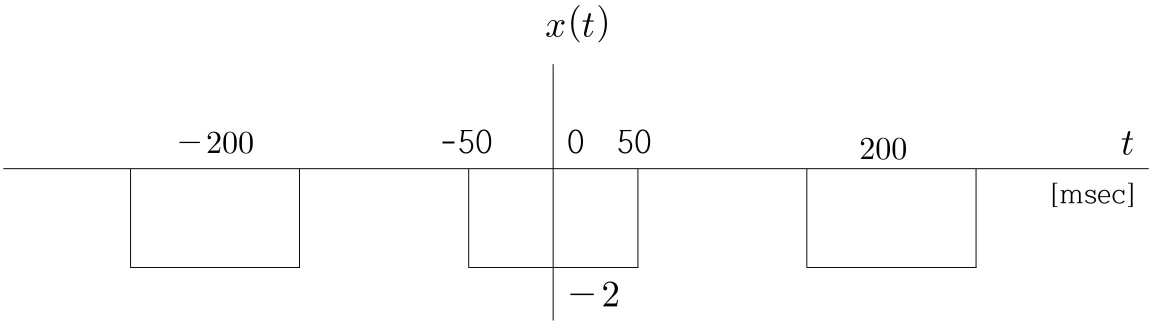
   the spectrum is a continuous function of ω.

[10-14] The spectrum of a periodic square waveform is given. The period of the square wave is 100 [msec].



10. Find the amplitude of the periodic square waveform.
11. Find the duty cycle [%] of the periodic square waveform.
12. Find the pulse width [msec] of the periodic square waveform.
13. Find the fundamental frequency [Hz] of the periodic square waveform.
14. Find the first zero-crossing frequency [Hz] of the spectrum.

[15-25] A periodic square wave $x(t)$ is given. (Period T= 200 msec, Amplitude= -2)



15. Find the fundamental frequency [Hz].

16. Find the first zero crossing frequency [Hz].

17. Find the frequency [Hz] of the spectrum interval?

18. Find the average value of this signal?

19. Find the Fourier coefficient, a_k .

(1) $\frac{4}{k\pi}\sin(\frac{k\pi}{4})$ (2) $\frac{1}{k\pi}\sin(\frac{k\pi}{2})$ (3) $-\frac{4}{k\pi}\sin(\frac{k\pi}{2})$

(4) $-\frac{2}{k\pi}\sin(\frac{k\pi}{2})$ (5) $\frac{2}{k\pi}\sin(k\pi)$ (6) $-\frac{4}{k\pi}\sin(\frac{k\pi}{4})$

20. Find a_0 .

(1) -1 (2) $-\frac{1}{\pi}$ (3) -2 (4) $\frac{1}{\pi}$ (5) $\frac{1}{4}$ (6) 0

21. Find a_1 .

(1) $-\frac{\sqrt{2}}{2\pi}$ (2) $-\frac{2}{\pi}$ (3) 0 (4) $\frac{4}{\pi}$ (5) $\frac{\sqrt{2}}{2\pi}$ (6) $\frac{2}{\pi}$

22. Find a_2 .

(1) $-\frac{\sqrt{2}}{2\pi}$ (2) $\frac{\sqrt{2}}{\pi}$ (3) $-\frac{1}{\pi}$ (4) $\frac{2}{\pi}$ (5) $\frac{1}{2\pi}$ (6) 0

23. Find a_3 .

(1) $-\frac{\sqrt{2}}{3\pi}$ (2) 0 (3) $\frac{\sqrt{2}}{6\pi}$ (4) $\frac{\sqrt{2}}{3\pi}$ (5) $\frac{2}{3\pi}$ (6) $-\frac{\sqrt{2}}{6\pi}$

24. Find a_4 .

(1) 0 (2) $-\frac{1}{2\pi}$ (3) -0.5 (4) $-\frac{2}{\pi}$ (5) $\frac{2}{\pi}$ (6) 0.5

25. Find a_6 .

(1) $-\frac{\sqrt{2}}{3\pi}$ (2) $-\frac{4}{3\pi}$ (3) 0 (4) $\frac{\sqrt{2}}{3\pi}$ (5) $\frac{2}{3\pi}$ (6) $\frac{4}{3\pi}$